

ESSENTIAL SKILL

Fire & Heat

Warmth. Cooking. Hot Water.

Fire can become one of our most useful resources when normal power and heating systems stop working.

- Warmth.
- Hot water.
- Cooking.
- A way to boil water for purification.

CREATE HEAT WITHOUT CREATING ANOTHER EMERGENCY.

IMMEDIATE ACTION

If you need heat right now

1

Keep people warm before trying to heat the whole building.

Use warm clothing, hats, gloves, blankets, sleeping bags, and layers.

2

Choose one smaller room or area.

It usually takes much less fuel to keep one occupied area warm than an entire building.

3

Use only heating equipment designed for the location where you are using it.

4

Maintain ventilation when required.

5

Have working smoke and carbon-monoxide alarms.

INVISIBLE DANGER

Carbon monoxide

You cannot see it.

You cannot smell it.

Carbon monoxide can cause headache, dizziness, weakness, nausea, confusion, unconsciousness, and death.

If a carbon-monoxide alarm sounds or people suddenly develop these symptoms around combustion equipment, get everyone into fresh air and seek emergency assistance.

KEEP COMBUSTION OUTSIDE

Never use these indoors

✗ Charcoal or barbecue grills.

✗ Portable gasoline generators.

✗ Gasoline engines.

✗ Outdoor camp stoves not approved for indoor use.

✗ A gas oven or kitchen range used to heat a building.

20 feet

Operate a generator outdoors, at least this far from windows, doors, and vents.

Using a fireplace or wood stove

Use only equipment designed and installed for indoor heating. Before depending on it:

☐ Know that the chimney or flue is clear.

☐ Make sure the stove or fireplace is in good condition.

☐ Have a fire extinguisher nearby.

☐ Have working smoke and carbon-monoxide alarms.

☐ Keep combustible materials away from the fire.

☐ Know how to control and extinguish the fire.

Chimneys should be inspected and cleaned regularly. Blocked chimneys can allow dangerous carbon monoxide to accumulate indoors.

FUEL PREPARATION

Wood

If wood is available locally, it can provide a renewable source of emergency heat and cooking fuel. But having trees is not the same as having usable firewood.

You need a way to cut, split, move, and store it, as well as people who know how to work safely.

Basic wood tools



Bow saw or hand saw



Axe



Splitting maul



Wedges



Work gloves



Eye protection



Sharpening equipment



Tarp or covered storage

Power saws can make the work faster, but hand tools remain useful when fuel or electricity is unavailable.

Dry wood matters

Wet wood is difficult to ignite, produces less useful heat and more smoke, and can create more deposits in chimneys.

Keep dry firewood protected from rain and snow. Do not wait for an emergency to begin cutting all the wood you will need.

BUILD GRADUALLY

How to start a basic wood fire

1

Tinder

Material that catches easily:
dry paper, dry grass, fine
dry wood shavings, or
commercial fire starters.

2

Kindling

Small dry sticks and pieces
of wood.

3

Firewood

Larger pieces that burn
longer and provide heat.

Tinder → Kindling → Small Wood → Larger Wood

Do not smother a small flame by adding large logs too quickly.

You also need air

Heat + Fuel + Oxygen

If a fire repeatedly goes out, one of those three things is usually missing. Good airflow helps a fire burn efficiently, but always use indoor equipment as designed so smoke and sparks stay out of the room.

CONTROLLED HEAT

Cooking with fire

An emergency cooking system does not need to be complicated. The goal is simply to create controlled heat beneath a pot.

- A wood stove with a cooking surface.
- A properly installed cookstove.
- An outdoor rocket stove.
- An outdoor grill or safe outdoor cooking fire.

Outdoor cooking equipment stays outdoors. Never move a charcoal grill or ordinary camping stove indoors because the weather is bad.

CONNECTED SYSTEMS

Fire and water

A dependable heat source allows you to boil questionable water to kill germs. An emergency water plan should answer:

How would I boil water if the electricity and gas were both off?

A large pot and a dependable alternative heat source can be extremely valuable.

RECOMMENDED SUPPLIES

Simple fire & heat kit

☐ Dry firewood

☐ Matches and lighters

☐ Fire starters

☐ Hand saw

☐ Axe or splitting maul

☐ Work gloves

☐ Large cooking pot

☐ Safe heating or cooking appliance

☐ Fire extinguisher

☐ Smoke alarms

☐ Carbon-monoxide alarms

☐ Flashlights or headlamps

☐ Warm clothing

☐ Blankets or sleeping bags

AVOID THESE

Common mistakes

- ✗ Use outdoor combustion equipment indoors.
- ✗ Run a generator in a house, basement, garage, or carport.
- ✗ Put a generator near doors or windows.
- ✗ Use a kitchen oven to heat the house.
- ✗ Burn wood in an unvented indoor space.
- ✗ Leave a fire unattended or allow children near an unsupervised fire.
- ✗ Store gasoline beside a stove or fireplace.
- ✗ Assume opening a window makes unsafe equipment safe.
- ✗ Wait until an emergency to learn how equipment works.

TRY THIS NOW

Fire & heat practice

Pick the emergency heating or cooking system you would actually depend upon. Then answer:

1. Where is it, and does it work?
2. Do I know how to operate it?
3. Where is the fuel, and how much do I have?
4. Can I safely boil a large pot of water with it?
5. Are working smoke and carbon-monoxide alarms nearby?
6. Where is the fire extinguisher?

Then actually practice using the system safely.

Do not discover how it works during the emergency.

BEFORE AN EMERGENCY

Quick checklist

☐ I know how I would stay warm without electricity.

☐ I have warm clothing and blankets.

☐ I have a safe alternative heat source and know how to operate it.

☐ I have fuel available.

☐ I have basic wood-cutting tools if I depend on wood.

☐ I have a way to boil water.

☐ My chimney or flue is safe.

☐ I have working smoke and carbon-monoxide alarms.

☐ I have a fire extinguisher.

REMEMBER

Warm the people before trying to warm the building.

Never bring outdoor combustion equipment indoors.

Fire needs fuel, heat, and air, but people need clean air too.

USE FIRE. RESPECT FIRE. CONTROL FIRE.

Official guidance

Follow local emergency instructions and the manufacturer's directions for every heating, cooking, or power device.

CDC: Carbon Monoxide Poisoning Basics(<https://www.cdc.gov/carbon-monoxide/about/index.html>)

CDC: Protect Yourself During a Power Outage (<https://www.cdc.gov/natural-disasters/response/what-to-do-protect-yourself-during-a-power-outage.html>)

U.S. Fire Administration: Heating Fire Safety(<https://www.usfa.fema.gov/downloads/pdf/publications/heating-fire-safety-handout.pdf>)
